

Velocity and acceleration

THESE QUANTITIES TELL US HOW QUICKLY SOMETHING IS MOVING.

When motion of an object changes in rate or direction, the motion is described in terms of velocity and acceleration.

Speed and velocity

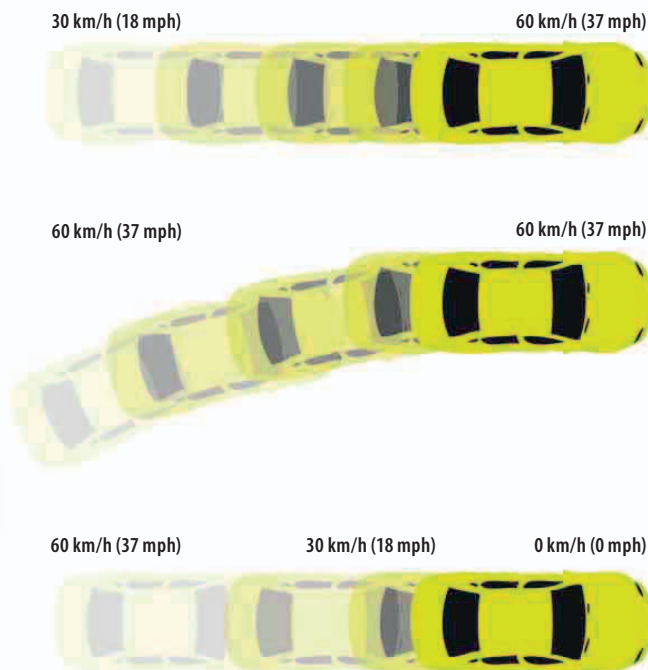
Speed is a measure of the rate at which a distance is covered. It is commonly measured in kilometers per hour. Velocity is also measured in these units, however, this measure also takes direction into account. Thermodynamicists and nuclear physicists use speed more often than velocity.

rate of motion (km/h)

space between two points on a straight line (km)

time that has elapsed between two points (hours)

$$\text{Velocity} = \frac{\text{Distance}}{\text{Time}}$$



Increasing velocity

An increase in the car's velocity is known as acceleration. A constant force gives a constant increase in velocity.

Changing direction

The car continues at 60 km/h (37 mph), but then changes lanes. The car's speed is constant, but its velocity is changing.

Decreasing velocity

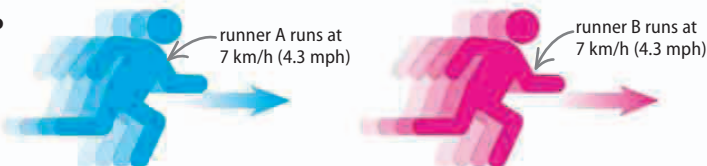
When a car slows, the change in velocity is known as a deceleration.

Relative velocity

The relative velocity compares how fast an object is traveling in comparison to another. If two objects are traveling in the same direction, the relative velocity can be calculated by subtracting the velocity of the slower object from the velocity of the quicker one. Two objects moving in the opposite direction along the same path would be heading for a collision. The relative velocity of these objects would be greater than either of their individual speeds.

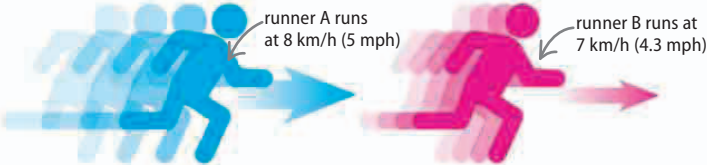
Relative speed zero

Runner A is moving at the same velocity as B, so their relative velocity is 0 km/h (0 mph).



Catching up

Runner A is gaining on B because his velocity is 1 km/h (0.6 mph) faster.



Heading for collision

Runner A and B are moving in opposite directions. Their relative velocity is the sum of their two speeds, which is 14 km/h (8.7 mph).

